PHYS551: Advanced Monte Carlo Methods

Homework 4

Student:
Ömer Faruk Avci

Instructor:
Prof. Dr. Taylan Akdoğan

Department of Electrical and Electronics Engineering

March 25, 2026

Disclosures and References

AI Usage Disclosure: In accordance with the course policy, I would like to declare the extent of AI assistance used in this assignment:

- **Analytical Derivations:** All mathematical derivations were performed manually without the use of AI tools.
- **Code Implementation:** The code was developed by me. AI tools were consulted to verify correctness and refine code comments.
- **Report Writing:** The explanations and discussions were written by me. AI tools were used to check for grammatical errors and to give feedback on the overall structure.

Problem 1.

$S = \{1, 2, 3\}$, $S_1 = 1$, $S_2 = 2$, $S_3 = 3$.

a).

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/5 & 1/2 & 3/10 \\ 0 & 2/5 & 3/5 \end{pmatrix}$$

row verification

$\frac{1}{2} + \frac{1}{2} + 0 = 1$ ✓

$\underset{(2)}{\frac{1}{5}} + \underset{(5)}{\frac{1}{2}} + \frac{3}{10} = 1$ ✓

$0 + \frac{2}{5} + \frac{3}{5} = 1$ ✓

b) $\pi P = \pi$ with $\pi_1 + \pi_2 + \pi_3 = 1$.

eq 1: $\frac{1}{5}\pi_1 + \frac{1}{5}\pi_2 + 0 \cdot \pi_3 = \pi_1 \longrightarrow \pi_1 = \frac{2}{5}\pi_2$

eq 2: $\frac{1}{2}\pi_1 + \frac{1}{2}\pi_2 + \frac{2}{5}\pi_3 = \pi_2$

eq 3: $0\,\pi_1 + \frac{3}{10}\pi_2 + \frac{3}{5}\pi_3 = \pi_3 \longrightarrow \pi_3 = \frac{3}{4}\pi_2$

$\pi_1 + \pi_2 + \pi_3 = 1$

$\underset{(4)}{\frac{2}{5}}\pi_2 + \underset{(20)}{\pi_2} + \underset{(5)}{\frac{3}{4}}\pi_2 = 1$ $\quad \frac{43}{20}\pi_2 = 1$ $\quad \boxed{\pi_2 = \frac{20}{43}}$ #

$\boxed{\pi_1 = \frac{2}{5} \cdot \frac{20}{43} = \frac{8}{43}}$ #

$\boxed{\pi_3 = \frac{3}{4} \cdot \frac{20}{43} = \frac{15}{43}}$ #

c) Checking all pairs for $\pi_i P_{ij} \overset{?}{=} \pi_j P_{ji}$

pair (i,j)	$\pi_i P_{ij}$	$\pi_j P_{ji}$	Check
$(1,2)$	$\frac{8}{43} \cdot \frac{1}{2} = \frac{4}{43}$	$\frac{20}{43} \cdot \frac{1}{5} = \frac{4}{43}$	✓
$(1,3)$	$\frac{8}{43} \cdot 0 = 0$	$\frac{15}{43} \cdot 0 = 0$	✓
$(2,3)$	$\frac{20}{43} \cdot \frac{3}{10} = \frac{6}{43}$	$\frac{15}{43} \cdot \frac{2}{5} = \frac{6}{43}$	✓

Q.E.D.

all three satisfied detailed balance holds.

Problem 2: Mixing Time and The Spectral Gap (Computational)

Part (a): Simulating the Evolution

Starting from $\mu^{(0)} = [1, 0, 0]$, we compute $\mu^{(t)} = \mu^{(0)} P^t$ by iteratively multiplying by P.

Python Source Code for Problem 2

```python
"""

# PHYS 551: Advanced Monte Carlo Methods
# Homework 3 - Source Code
# Student: Omer Faruk Avci
# Instructor: Prof. Dr. Taylan Akdogan

"""
import numpy as np
import matplotlib.pyplot as plt

"""Problem 2: Mixing Time and The Spectral Gap (Computational)"""


# Transition matrix
P = np.array([
    [1/2, 1/2,   0  ],
    [1/5, 1/2, 3/10 ],
    [0,   2/5,  3/5 ]
])

# Initial distribution given in the problem
mu_init = np.array([1.0, 0.0, 0.0])

t_max = 40
mu_values = np.zeros((t_max + 1, 3))
mu_values[0] = mu_init

for t in range(1, t_max + 1):
    mu_values[t] = mu_values[t-1] @ P  # mu^(t) = mu^(0) * P^t

# Part b: Plot
pi = np.array([8/43, 20/43, 15/43])
time = np.arange(t_max + 1)

plt.figure(figsize=(10, 6))
colors = ['red', 'orange', 'blue']
for i, c in enumerate(colors):
    plt.plot(time, mu_values[:, i], f'-o', color=c, label=f'State {i+1}')
    plt.axhline(pi[i], color=c, linestyle='--', alpha=0.6)

plt.xlabel('Time')
plt.ylabel('Probability')
plt.title('Markov Chain Evolution')
plt.legend(); plt.grid(True, alpha=0.6)
plt.tight_layout()
plt.savefig("markov_evolution.png")

# Part c: Eigenvalues
eigenvalues = np.linalg.eigvals(P)
eigenvalues_sorted = sorted(eigenvalues, key=abs, reverse=True)

lambda_1 = eigenvalues_sorted[0]
lambda_2 = eigenvalues_sorted[1]
gamma    = 1 - abs(lambda_2)
tau      = -1 / np.log(abs(lambda_2))

print(f"lambda1 = {lambda_1:.6f}")
print(f"lambda2 = {lambda_2:.6f}")
print(f"Spectral Gap = {gamma:.6f}")
print(f"Relaxation Time = {tau:.6f}")
```

Part (b): Visualization

The figure below shows $\mu^{(t)}$ for each state as a function of $t \in [0, 40]$. The horizontal dashed lines represent the exact analytical stationary values found in Problem 1b.

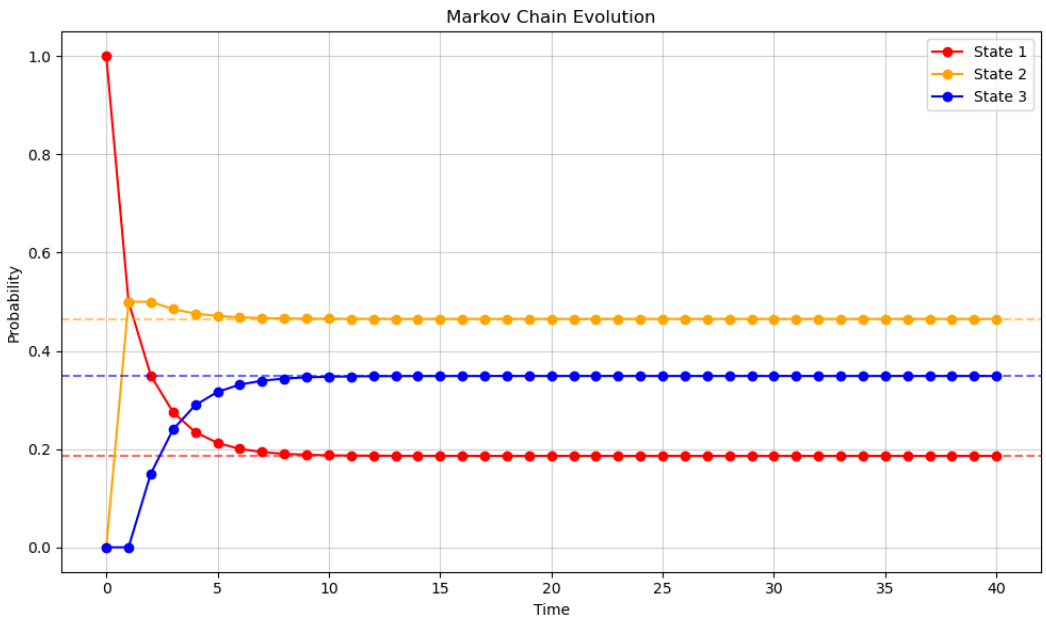

Figure 1: Markov Chain Evolution

The chain converges visually to the stationary distribution by approximately $t \approx 8$–10, consistent with the spectral gap analysis below.

Part (c): Spectral Gap and Relaxation Time

The three eigenvalues of P, sorted by absolute value, are:

Quantity	**Value**
Largest eigenvalue λ_1	1.000000
Second largest eigenvalue $\lvert\lambda_2\rvert$	0.544949
Spectral Gap $\gamma = 1 - \lvert\lambda_2\rvert$	0.455051
Relaxation time $\tau = -1/\ln\lvert\lambda_2\rvert$	1.647275

Table 1: Eigenvalue analysis and derived mixing quantities.

Terminal Output

```
lambda 1 = 1.000000
lambda 2 = 0.544949
Spectral Gap = 0.455051
Relaxation Time = 1.647275
```

Interpretation of τ: The total variation distance between $\mu^{(t)}$ and π decays approximately as:

$$\|\mu^{(t)} - \pi\|_{\mathrm{TV}} \propto e^{-t/\tau}$$

At $t = \tau \approx 1.65$ steps, the distance has already decreased by a factor of $1/e$. In practice, convergence is accepted when $t \approx 5\tau$:

$$5\tau \approx 8.24 \text{ steps}$$

This is fully consistent with the plot in Part (b), where all three state probabilities have settled onto their stationary dashed lines by $t \approx 8$–10.